QUANT

Mensuration & Geometry

(Triangles, Polygons & Mensuration)

Assignment 02

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Geometry Online Assignment Cluster I -02

1. If the area and perimeter of a rectangle are 240 m^2 and 68 m respectively, find its length.

1. 12 m 2. 24 m 3. 48 m 4. None of these

2. In a trapezium, the length of parallel sides are 20 and 25 metres respectively and the perpendicular distance between the parallel sides is 12 metres. Find the area of trapezium.

1. 540 m^2 2. 300 m^2 3. 450 m^2 4. 270 m^2

3. A rectangular plot 15 m × 10 m has a path of grass outside it. If the area of grassy pathway is 54 m^2, find the width of the path.

1. 1 m 2. 1.5 m 3. 2 m 4. 2.5 m

4. A square carpet with an area 169 m^2 must have 2 metres cut off one of its edges in order to be a perfect fit for a rectangular room. What is the area of rectangular room?

1. 195 sq. m 2. 55 sq. m 3. 26 sq. m 4. 143 sq. m

5. The radius of a sphere is increased by 50 %. The increase in the surface area of sphere is

1. 100% 2. 125% 3. 150% 4. 200%

6. The area of the four walls of a room is 1,080 square metres. The height and length of the room are in the ratio of 2 : 5. The height and breadth are in the ratio of 4 : 5. The carpet area of the room is

1. 450 m^2 2. 1800 m^2 3. 900 m^2 4. 300 m^2

7. If the side of a square is increased by 200 %, the area of the square increases to

1. 200% 2. 800% 3. 900% 4. 300%

8. If the of circle is increased by 20 %, the area of the circle increases by

1. 40% 2. 400 % 3. 44% 4. 87%

9. If the length of a rectangle is increased by 30% and breadth by 20 %, the area of the rectangle increases by

1. 80% 2. 56% 3. 50% 4. None of these

10. Volume of cylinder is decreased by 36 % when its radius is decreased by k%. find the value of k?

1. 80% 2. 20% 3. 90% 4. None of these

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Answer key & Explanations

Q. No.	Ans.	Exp.
1.	2	$lb = 240$, $2(l + b) = 68$, $l + b = 34$ $= 24 + 10$, $l = 24$ m.
2.	4	Area of trapezium = ½ × (sum of parallel sides) × perpendicular distance between them = ½ × (20 + 25) (12) = 270 m^2.
3.	1	Area of rectangle = 150 m^2, area of grass path = 54, area of inner rectangle = 204 m^2 = 17 × 12 = (15 + 2) × (10 + 2) Width of grassy path = 1 m.
4.	4	Length of side of square carpet = 13m. Area cut off = 13 × 2 = 26 m, Area of room = 13 × 11 = 143 m^2.
5.	2	Surface area of sphere = $4\pi r^2$, if radius becomes $1.5r$ then surface area = $4\pi(1.5 \times r)^2 = 2.25(4\pi r^2)$, % increase in surface area = $[2.25 - 1](4\pi r^2) / 4\pi r^2 \times 100 = 125\%$.
6.	1	Height : Length = 2 : 5 , height : breadth = 4 : 5, height : length : breadth = 4 : 10 : 5, area of four walls of room , $1080 = 2[(10 + 5)k]\,4k$ $k^2 = 9$, $k = 3$. Carpet area = lb = (10 × 3) (5 × 3) = 450 sq m.
7.	2	If side of square increase by 200% is it becomes 3 times. Then the area becomes 9 times. Area increases to 900% of initial and by 800% of initial.
8.	2	If radius of circle is increased by 20% Area becomes 1.44 times. Hence it is increased by 44%.
9.	2	If the length of a rectangle is increased by 30% and breadth by 20 % area becomes 1.3 x1.2=1.56 times .hence it increases by 56%.
10.	2	Original volume = $\pi r^2 h$ New volume= $.64\,\pi r^2 h$ $= \pi(.8r)^2 h$ Hence radius decreases by 20%